Task 4 EEG waveform decoding

Initial baseline, event audit, and interpretation limits

Satyam Awasthi

Revised 20 October 2026 | Analysis run: 16 October 2026

Main finding. The fixed mean-amplitude baseline did not demonstrate reliable five-class waveform decoding: mean within-participant balanced accuracy was 19.1%, and pooled leave-one-subject-out balanced accuracy was 18.5%, against a 20% chance reference. This finding applies to this minimally processed pipeline; it does not establish that waveform information is absent.

1 Question and scope

Can one trigger-aligned, post-stimulation EEG epoch identify which of five median-nerve TENS waveforms was delivered? The immediate target is *delivered waveform identity*, not the participant's naturalness preference. The wider motivation is understanding waveform-dependent sensation for sensory restoration. Collu et al. demonstrated perceptual differences between some charge- and amplitude-matched shapes [1]; the present baseline asks a different question about EEG prediction.

This note summarizes the first completed Task 4 experiment, including its negative primary result and diagnostic controls. Detailed channel-quality, evoked-response, and feature/covariance investigations remain planned work. No Task 3 classifier was run for this note.

2 Data and event audit

The cohort contains 10 participants, subjects 3–12, recorded with the SAGA system at 1,000 Hz. Subject 2 is excluded from Task 4 because Jason Whitson confirmed an incompatible earlier task version. Jason's corrected table is the dataset label reference, confirmed through project correspondence: Sine=239, Square=240, Triangle=241, linear decrease=242, and linear increase=243. Task 4 CSV waveform numbers 4 and 5 mean linear decrease and increase, respectively. Historical thesis task numbers differ from dataset task numbers [2].

Each Task 4 comparison presents two waveforms, with within-pair order varying across trials. A comparison ends when one waveform receives five preference selections (race to five). Jason advises at most 1.2 s post-stimulus for Task 4 epochs because the two stimuli are approximately 1.4 s apart.

Audit outcome	Result
Debounced events matching the complete CSV sequence	1,246
Events eligible after pair-level EEG quality exclusion	1,244
EEG pairs excluded by the fixed amplitude criterion	1
Observed within-pair interval range (s)	1.408–1.448
Comparison blocks inconsistent with race to five	0

3 Baseline methods

3.1 Extraction, alignment, and eligibility

Our loader selects the single XDF stream containing both EEG and `TRIGGERS`, verifies EEG units, and converts microvolts to volts. Recorded timestamps are retained without clock correction or dejittering. Expected trigger values must remain constant for at least three samples. Accepted events retain the original run-start sample; short runs, unexpected codes, and active-low normalization are logged. For example, subject 3's accepted runs lasted eight or nine samples; its two unexpected runs were rejected. This is a digital-event audit, not an independent current-onset check.

The full accepted sequence must exactly match the CSV's first/second waveform columns in original row order. No forward scanning or automatic ambiguous resynchronization is used. All participants passed this check. Valid current responses are values 1 or 2 in `T4 Naturalness Value`. All matched current choices were valid in this cohort. A missing response would not by itself prove that current delivery failed.

Epochs extending beyond recording boundaries, nonfinite EEG samples, timestamp irregularities exceeding 10% of the nominal sample step, and neighboring stimulation triggers within the required epoch span are flagged. A fixed 500 μV peak-to-peak threshold is checked over selected referenced channels in the non-onset pre/post spans. If one presentation fails, both members of its pair are excluded from primary evaluation; all original rows and reasons remain in the manifest.

3.2 Features and fixed classifier

CREF, M1, and M2 are removed, then a fixed spatial average reference is applied over remaining EEG channels. Features use 14 predefined electrodes: FC1, FC2, C3, C1, Cz, C2, C4, CP5, CP1, CP2, CP6, P3, Pz, and P4. Each feature is a channel's bin mean minus its specified baseline mean.

Features	Interval (ms)	Bin (ms)	Baseline (ms)	Dimension
Post (primary)	50–500	50	−250 to −50	126
Pre (control)	−500 to −50	50	−750 to −550	126
Late (sensitivity)	150–500	50	−250 to −50	98
Onset (diagnostic)	0–20	5	−250 to −50	56
Order/history	No EEG	–	–	10

Intervals are half-open. Pre and post feature windows have equal duration and separate preceding 200 ms baselines. Order/history features contain elapsed recording time, event ordinal, first/second position, preceding-event gap, and six indicators for the preceding trigger (including no preceding event). They exclude the current trigger and comparison identity.

This initial baseline did not run the supplied preprocessing pipeline. No temporal filtering, resampling, ICA, artifact interpolation, or source localization was applied. Thus post-onset activity cannot be smeared backward by a temporal filter in this pipeline. This deliberate simplification does not remove drift, line noise, blinks, muscle activity, or electrical contamination. Every model uses training-fold standardization followed by LDA with automatic covariance shrinkage, the least-squares solver, and equal class priors. There is no hyperparameter search or data-driven channel selection.

3.3 Validation and statistical summaries

Within participants: five group folds hold out entire unordered waveform comparisons. Both presentations of a trial stay together, and repeated occurrences of a comparison share a group. Consequently, this evaluation tests generalization to unseen *pair contexts*, not merely to randomly held-out trials. All training folds must contain all five classes.

Across participants: leave-one-subject-out (LOSO) evaluation holds out each complete participant, fitting standardization and LDA only on the remaining participants. The same eligible trials and outer splits are used for every feature condition. Within-participant training pairs are purged if their $[-750, +500]$ ms signal support overlaps a held-out event's support.

Balanced accuracy is five-class macro recall over pooled held-out predictions:

$$\mathrm{BA} = \frac{1}{5}\sum_{c=1}^{5}\frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FN}_c}.$$

All five classes occur in the complete evaluation set. Individual held-out comparison folds can contain fewer than five test classes; their mean fold-wise balanced accuracy is therefore not used as the headline metric. For participant s, let $\mathrm{BA}_s = \frac{1}{5}\sum_{c=1}^{5}\mathrm{Recall}_{s,c}$. The within-participant summary gives each participant equal weight:

$$\text{Mean within-participant BA} = \frac{1}{10}\sum_{s=1}^{10}\mathrm{BA}_s.$$

Pooled LOSO BA instead gives each class equal weight across all held-out events. These are different summaries, not a paired model comparison.

A descriptive 95% interval is estimated by a **participant-level bootstrap**: participants are resampled 10,000 times with replacement (seed 20261016).

We performed 99 restricted label randomizations (seed 42), preserving participant, waveform-comparison, and paired-trial structure. Labels were not freely shuffled across individual presentations. Within each comparison, whole ordered pairs of labels were circularly shifted, and first/second label orientation could be reversed for the complete comparison. Models were refitted using the same outer validation splits. These randomizations provide a diagnostic comparison; their interpretation depends on the validity of shifting labels within each comparison, which recording drift can limit.

Let $\mathrm{BA}_{\mathrm{obs}}$ be balanced accuracy using the true labels and $\mathrm{BA}_{\mathrm{null}}^{(b)}$ the balanced accuracy from randomized run b. The reported upper-tail quantity is

$$p = \frac{1 + \#\{b : \mathrm{BA}_{\mathrm{null}}^{(b)} \geq \mathrm{BA}_{\mathrm{obs}}\}}{99 + 1}.$$

Here # counts how many of the 99 randomized runs matched or exceeded the observed score. The added 1 and denominator $99 + 1 = 100$ provide the finite-randomization correction, avoiding a reported p-value of zero. Individual-participant randomizations are not pooled into a cohort test.

4 Initial findings

Table 1 and Figure 1 summarize the predefined conditions. Primary within-participant BA was 19.1%, with a descriptive participant-bootstrap interval of 15.7–22.4%. Primary pooled LOSO BA was 18.5%, with restricted-permutation $p = 0.87$. These results do not support reliable above-chance decoding under the tested pipeline.

Table 1: Five-class prediction results. Accuracy entries are percentages; F1 is on a 0–1 scale. All rows use 1,244 eligible events.

Features	Mean within BA	Pooled LOSO BA	LOSO macro-F1
Post-stimulus (primary)	19.1	18.5	0.182
Pre-stimulus	16.8	19.3	0.193
Late window	21.9	19.8	0.195
Onset	16.1	20.7	0.205
Order/history only	10.5	19.5	0.177

The late window's mean within-participant score (21.9%) is an exploratory sensitivity result; it is not evidence of reliable cohort-level decoding. Primary LOSO BA was 18.4% for first presentations and 18.5% for second presentations (622 events each). These descriptive values do not constitute a powered test of position effects. Full position-specific responses and feature distributions remain to be inspected. Participant-level primary post-stimulus results, together with pre-stimulus control scores, are provided in Appendix A.

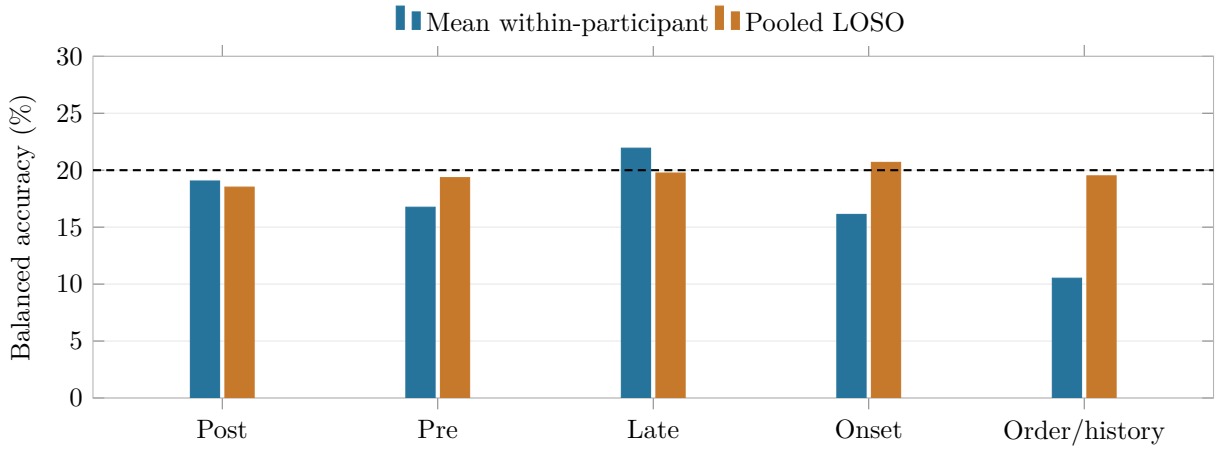


Figure 1: Predefined feature/control conditions. The dashed line is the 20% chance reference, not a significance threshold. Bars show point summaries only; uncertainty for the primary within-participant mean is reported in the text. The two evaluation settings use different averaging schemes.

5 Interpretation and limitations

The primary result is negative for this baseline. The selected features and classifier did not show easy waveform separation under grouped validation. This does not establish that EEG lacks waveform information or that more appropriate processing cannot recover it. No feature/model choice was optimized after observing the scores reported here.

Controls do not prove cleanliness. Pre-stimulus decoding is not systematically stronger than the primary window in the cohort summary. Its slightly higher pooled LOSO value remains near chance. Low pre scores do not rule out post-only electrical artifacts; onset or late scores alone cannot distinguish electrical from neural information. Although this pipeline avoids temporal-filter smearing, prior-stimulus responses and block-related state can still affect both signal windows and baselines.

The within-participant order/history score is low under held-out comparison contexts: a relation learned in one waveform pair may fail in another. It is not evidence that acquisition order is uninformative under other splits. Because preceding-trigger indicators are label-related history, the restricted permutation distribution for that control is a diagnostic comparator rather than a clean physical randomization of acquisition order.

Digital alignment is not physical timing calibration. Exact CSV/XDF sequence agreement establishes consistency of recorded codes and order. The thesis describes triggers sent at waveform start and a current-monitor feedback path. Jason confirmed that the monitor traces were saved on the collection laptop, but those traces are not part of this analysis. Hardware synchronization and trigger-to-current delay/jitter remain unverified.

Potential charge/intensity confounding. Task 4 used waveform-specific detection thresholds, and equal delivered charge across waveform conditions was not established. Any future above-chance decoding could therefore reflect charge or intensity differences rather than waveform shape alone.

References

[1] R. Collu, E. J. Earley, M. Barbaro, and M. Ortiz-Catalan. Non-rectangular neurostimulation waveforms elicit varied sensation quality and perceptive fields on the hand. *Scientific Reports*, 13:1588, 2023. doi:10.1038/s41598-023-28594-0.

[2] J. Whitson. *Beyond the Square Pulse: Waveform Shape, EEG Correlates, and the Pursuit of Natural Sensation in TENS*. Honors Undergraduate Thesis, University of Central Florida, 2026. `https://stars.library.ucf.edu/hut2024/661/`.

Appendix A Participant-level primary results

Table 2: Deidentified participant-level results. Within scores hold out comparison contexts; LOSO scores are predictions when that participant is held out entirely. All accuracy entries are percentages.

Participant	Eligible events	Within post BA	Within pre BA	LOSO post BA
S03	114	22.4	20.1	13.3
S04	122	18.0	22.4	17.3
S05	142	25.3	17.4	16.5
S06	122	20.3	12.4	14.5
S07	116	27.8	12.8	19.9
S08	130	17.4	18.5	20.7
S09	136	13.1	20.8	19.0
S10	114	23.0	14.8	21.2
S11	116	11.3	11.2	16.4
S12	132	12.0	17.2	24.9

Participant-specific variation is substantial. Uncorrected exploratory participant/control findings are not treated as discoveries; there are multiple participants and multiple diagnostic windows. The 99-permutation resolution is 0.01, and the restricted nulls require their own exchangeability assumptions. The main conclusion is limited to absence of reliable cohort-level performance in this first fixed baseline.

Raw EEG, behavioral CSVs, and participant-level predictions are not embedded in this PDF. This appendix contains deidentified accuracy summaries only; the analysis configuration and original run records are retained separately.